

# Coupling Electronic Topology to Phonons and Magnetism in Semimetals

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A dissertation  
submitted to the Faculty of  
the department of Physics  
in partial fulfillment  
of the requirements  
for the degree of  
Doctor of Philosophy

Boston College  
Morrissey College of Arts and Sciences Graduate School

July 2026

PREVIEW

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## ABSTRACT

Semimetallic quantum materials provide a platform in which electronic structure, lattice dynamics, and magnetic order can become strongly intertwined. While many of these materials are introduced through symmetry-constrained band crossings, Berry curvature, or related band-structure features, their experimentally observed properties are shaped by phonons, spin textures, and the scattering processes that connect them to itinerant electrons. This dissertation investigates how phonons and magnetism couple to semimetallic electronic states and how these couplings appear in transport and scattering measurements.

The first part of the dissertation focuses on the nonmagnetic  $MX_2$  semimetal family, where  $M = \text{Nb}$  or  $\text{Ta}$  and  $X = \text{Ge}$  or  $\text{Si}$ . By combining electrical transport, quantum oscillations, Raman scattering, and first-principles calculations, this work examines the origin of anomalously large transport mobility. The results show that phonon-electron dominated scattering can enhance conductivity by returning momentum to the electronic system, establishing lattice dynamics as an active ingredient in charge transport.

The second part turns to magnetic Weyl semimetals, where electronic topology can couple directly to local-moment magnetism. In  $\text{SmAlSi}$ , neutron diffraction, magnetization, heat-capacity, transport, and electronic-structure analysis reveal spiral magnetic order, a field-induced A-phase, and a large topological Hall effect, suggesting

a mechanism in which topological Fermi pockets participate in stabilizing complex magnetic order. Across the  $\text{CeAlSi}_{1-x}\text{Ge}_x$  series, Ge-rich compositions develop incommensurate magnetism and singular angular magnetoresistance, providing an electrical fingerprint of the underlying magnetic texture.

Together, these studies show that topology in semimetals should not be viewed as an isolated band-structure property. Instead, electronic topology, phonons, and magnetism form coupled systems whose interactions produce measurable transport, magnetic, and spectroscopic responses.

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## ACKNOWLEDGEMENTS

I would like to express my deepest gratitude to my advisor, Prof. Fazel Tafti. I am grateful not only for his scientific guidance, but also for his kindness, patience, and constant encouragement throughout my Ph.D. He has always been generous with his time and ideas, and his support helped me grow as a researcher and as a person. His confidence in me made it possible for me to continue through difficult moments, and his way of thinking about physics shaped how I approach the problems in this dissertation.

I am also grateful to the members of my dissertation committee for their time, advice, and thoughtful feedback. Their questions and suggestions helped me see this work from a broader perspective and improved both the scientific content and the presentation of this dissertation.

I would like to give special thanks to previous members of the Tafti group, especially Hung-Yu Yang and Faranak Bahrami. They taught me a great deal when I first joined the lab, from practical experimental skills to the everyday habits needed to do careful research. More importantly, they were patient, kind, and helpful far beyond laboratory matters. Their support made the beginning of graduate school much less overwhelming.

I am thankful to the members of the Tafti group, past and present, for creating a supportive research environment and for sharing many discussions, measurements, and ordinary days in the lab. I would especially like to thank Kyle Fruhling, Alenna Streeter, Sudhaman Balguri, Piyush Sakrikar, Enrique Gonzalez, Siddharth Nan-

danwar, Taichi Murakami, and Jake Scally. I have learned from your questions, your experimental experience, your willingness to help, and the friendship that made graduate school feel warmer. I would also like to thank Xinyi Zhao especially, not only as a labmate but also as one of my closest friends in the Tafti group. Her warmth, kindness, and constant encouragement made many difficult days feel lighter.

I am also deeply grateful to my collaborators, especially Jonathan Gaudet and Predrag Nikolic, for many helpful discussions and close collaboration on the magnetic Weyl semimetal projects. Their insight helped me understand how neutron scattering, transport, and theoretical modeling can be brought together into one physical picture.

I am also grateful to the staff members at Boston College who helped me in many ways during my time here. I would like to thank Jane Carter, who cared about me when I was sick and helped me during my move, and Nancy Chervy, Scott Bortolotto, and Kiel Smith for their help, patience, and support. Their kindness and reliability made many practical parts of graduate life much easier.

I would like to thank my friends Yiping Wang, Yuxuan Wang, and Xin Li for their companionship, encouragement, and understanding. I am especially grateful to Xinyue Zhang, who was my roommate and close friend; our conversations, shared interests, and many ordinary moments together made life outside the lab warmer and happier. I am also especially grateful to Keyu Zeng, who was also my roommate at a different time and brought kindness, patience, and many good conversations into my daily life.

I would also like to recognize my dear friends Zhuqi Feng, Tong Bao, Qiushi Liang, Xiyuan Kang, and Kayoos. We came into each other's lives through different paths, including school, everyday life, animations and games, but your friendship has been a quiet and important source of support throughout these years. Our conversations, encouragement, and moments of mutual comfort helped me through many difficult days. Since one only writes a dissertation once, I am very happy to have your names

included here.

I am deeply thankful to my boyfriend, Zhenduo Wang, for his love, patience, and steady support through the long and sometimes stressful process of completing this degree. Over the four years we spent together in college and the seven years we have spent across different countries, we have continued to share a strong sense of connection and understanding. I am grateful for the many conversations we have had about physics, future work, difficult bottlenecks, everyday life, entertainment, and games. His companionship has helped me think more clearly, feel less alone in difficult moments, and find joy outside the pressure of graduate school.

Finally, I want to thank my parents, Yingyuan Yao and Chunxin Yang, for their unconditional love and support. They have always believed in me, even when I was far from home and uncertain about myself. This dissertation would not have been possible without their encouragement, sacrifice, and confidence in me.

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# CHAPTER I

## Introduction

Topological and topologically related semimetals have emerged as a central platform for studying electronic states whose properties are constrained by symmetry, spin-orbit coupling, and band geometry. Their band structures can host Dirac or Weyl crossings, Fermi-arc surface states, large Berry curvature, or other symmetry-constrained electronic features, leading to unusual responses to electric, magnetic, and optical fields. These features have motivated intense interest in semimetals both as fundamental quantum materials and as possible platforms for electronic, spintronic, and optoelectronic applications.

Much of the language used to describe topological semimetals begins from a single-particle band picture. In this view, the most important ingredients are the crystal symmetry, spin-orbit coupling, band inversion, and the location of topological band crossings relative to the Fermi level. This picture is essential, but it is not complete. Real materials contain phonons, defects, electronic correlations, local moments, magnetic domains, and structural imperfections. Electrons scatter from these degrees of freedom, exchange energy and momentum with them, and in some cases help stabilize them. As a result, the experimentally measured properties of topological semimetals can reflect not only the topology of the electronic structure, but also the way that topology is coupled to lattice vibrations, magnetic textures, and ordered states.

The central theme of this dissertation is that electronic structure should be studied together with the lattice and magnetic degrees of freedom to which it couples. In the materials discussed here, phonons and spin textures are not passive backgrounds. Phonons can control whether electronic momentum is relaxed or returned to the electron system. Magnetic order can be shaped by the topology and geometry of the electronic structure and can, in turn, generate unusual Hall and magnetoresistance responses. Transport measurements provide one of the most direct ways to observe these effects, but transport is treated here as a window into a broader physical question: how do semimetallic electrons, phonons, and magnetism form coupled quantum-material systems?

This dissertation addresses that question through three material platforms. The first is the nonmagnetic  $MX_2$  semimetal family, where  $M = \text{Nb}$  or  $\text{Ta}$  and  $X = \text{Ge}$  or  $\text{Si}$ . In this family, chemical substitution tunes the electronic and phonon structures while preserving the essential crystal structure, allowing transport to be compared with phonon and band-structure calculations across a controlled chemical series (Yao\* and Plisson\* et al., 2024). The second platform is  $\text{SmAlSi}$ , a magnetic Weyl semimetal in which spiral magnetic order and a large topological Hall effect point to a close relationship between Weyl electronic states and local-moment magnetism (Yao, Gaudet, et al., 2023). The third platform is the  $\text{CeAlSi}_{1-x}\text{Ge}_x$  series, where Ge substitution drives the system toward incommensurate magnetism and singular angular magnetoresistance as an electrical fingerprint of complex magnetic order (Yao, Chen, et al., 2026).

## 1.1 Topological Semimetals

Topological semimetals are characterized by band crossings near the Fermi level that are protected or constrained by crystal symmetry and topology. In Dirac semimetals, fourfold-degenerate Dirac points can occur when both time-reversal

and inversion symmetries, together with additional crystal symmetries, protect the crossing. In Weyl semimetals, either time-reversal or inversion symmetry is broken, and a Dirac point can split into pairs of Weyl nodes with opposite chirality. Near a Weyl node, the electronic dispersion is approximately linear, and the node acts as a source or sink of Berry curvature in momentum space (Hu et al., 2019).

The presence of Berry curvature gives Weyl semimetals a natural connection to anomalous transport phenomena. Depending on symmetry, magnetic order, and Fermi-level position, these materials can show unusual magnetoresistance, anomalous Hall responses, chiral-anomaly-related transport, and surface Fermi arcs. These effects make topological semimetals powerful systems for exploring the relationship between electronic structure and measurable response functions.

At the same time, topology alone rarely determines the full experimental behavior. Two materials with similar topological classifications can show very different mobilities, scattering rates, magnetic ground states, and field responses. This sensitivity reflects the importance of material-specific ingredients: carrier density, disorder, phonon spectra, magnetic exchange interactions, anisotropy, and Fermi-surface geometry. A major goal of this dissertation is to understand how these ingredients interact with topology rather than treating them as secondary complications.

## 1.2 Transport as a Probe of Coupled Degrees of Freedom

Electrical transport is one of the most widely used probes of topological semimetals because it is sensitive to carrier density, mobility, scattering, magnetic order, and Berry curvature. Longitudinal resistivity reveals how efficiently charge carriers move through a material, while Hall measurements provide information about carrier type, density, mobility, and anomalous contributions. Magnetoresistance and angular magnetoresistance further reveal how electronic motion depends on field strength and field direction.

In this dissertation, transport plays two related roles. In the  $MX_2$  compounds, transport is the central phenomenon to be explained: the measured transport mobility can be much larger than the single-particle mobility, suggesting that the processes limiting individual quasiparticle lifetimes do not necessarily relax the total electronic momentum. In  $\text{SmAlSi}$  and the  $\text{CeAlSi}_{1-x}\text{Ge}_x$  series, transport is not the only story. Instead, Hall and magnetoresistance measurements act as signatures of deeper coupling between electronic topology and magnetic order. This distinction is important for the overall logic of the thesis: transport is both a phenomenon in its own right and a diagnostic tool for coupled electronic, lattice, and magnetic behavior.

### 1.3 Electron-Phonon Coupling and Phonon Drag

In ordinary metals, electron-phonon scattering is often treated as a process that limits conductivity. An electron accelerated by an electric field scatters from a phonon, transfers momentum to the lattice, and thereby contributes to resistive dissipation. This picture is appropriate when phonons efficiently lose their momentum through phonon-phonon scattering, defects, or boundaries.

However, a different regime can occur when phonon-electron scattering is stronger than phonon-phonon scattering. In that case, a phonon that receives momentum from one electron can return that momentum to another electron before it decays into the broader phonon bath. The scattering of an individual electron may still be strong, but the total momentum of the coupled electron-phonon system can be partially preserved. This produces a phonon-drag-like enhancement of charge transport.

The  $MX_2$  chapter examines this physics directly. By comparing transport mobility, quantum mobility, Raman linewidths, and first-principles calculations across  $\text{NbGe}_2$ ,  $\text{TaGe}_2$ ,  $\text{NbSi}_2$ , and  $\text{TaSi}_2$ , the study tests whether phonon-electron dominated scattering can account for anomalously large transport mobility (Yao\* and Plisson\* et al., 2024). The key idea is that the phonon system can enhance, rather than

simply degrade, electronic transport when the dominant scattering channel returns momentum to the electron system.

## 1.4 Magnetic Weyl Semimetals

Magnetic Weyl semimetals introduce magnetic order as an additional active degree of freedom. In these systems, local moments and itinerant electronic states coexist, and their coupling can produce phenomena that are absent in nonmagnetic semimetals. Magnetic order can break time-reversal symmetry, reconstruct electronic bands, generate Berry curvature, and create anomalous Hall responses. Conversely, the electronic structure can influence the magnetic interactions between local moments.

The rare-earth aluminum silicides and germanides studied in this dissertation are especially useful for exploring this interplay. Their noncentrosymmetric crystal structures support Weyl electronic states, while the rare-earth ions provide local magnetic moments. This combination allows the topology of the electronic bands and the magnetism of the local moments to become intertwined. In SmAlSi, the spiral magnetic order is connected to the topology and nesting of the electronic structure (Gaudet et al., 2021; Yao, Gaudet, et al., 2023). In the CeAlSi<sub>1-x</sub>Ge<sub>x</sub> series, incommensurate magnetic order produces a singular angular magnetoresistance response, showing how transport can detect complex magnetic textures (Puphal et al., 2020; Yao, Chen, et al., 2026).

These materials therefore motivate a broader view of magnetic Weyl semimetals. Rather than treating magnetism only as a way to create Weyl nodes, one can ask how Weyl electrons participate in magnetic exchange, how magnetic textures reshape transport, and how transport measurements reveal the underlying spin structure. This viewpoint provides the basis for the SmAlSi and CeAlSi<sub>1-x</sub>Ge<sub>x</sub> chapters.

## 1.5 Dissertation Overview

Chapter II summarizes the experimental and computational methods used throughout the dissertation, including crystal growth, structural characterization, magnetotransport, Raman scattering, neutron diffraction, thermodynamic measurements, and first-principles calculations.

Chapter III focuses on electron-phonon coupling and enhanced transport in  $MX_2$  semimetals. This chapter asks how phonons can mediate momentum flow in a semimetal and shows that phonon-electron dominated scattering provides a mechanism for anomalously large transport mobility.

Chapter IV discusses topology-magnetism coupling in SmAlSi. The chapter centers on spiral magnetic order, its connection to topological electronic structure, and the large topological Hall effect that emerges as a transport signature of the magnetic texture.

Chapter V examines incommensurate magnetism and singular angular magnetoresistance across the  $\text{CeAlSi}_{1-x}\text{Ge}_x$  series. This chapter shows how angular magnetotransport can reveal complex magnetic order in a magnetic Weyl semimetal.

Chapter VI summarizes the unified physical picture developed across the dissertation. Together, the three projects show that semimetallic quantum materials are best understood as coupled systems in which electronic topology, lattice dynamics, and magnetism jointly shape the observed material response.